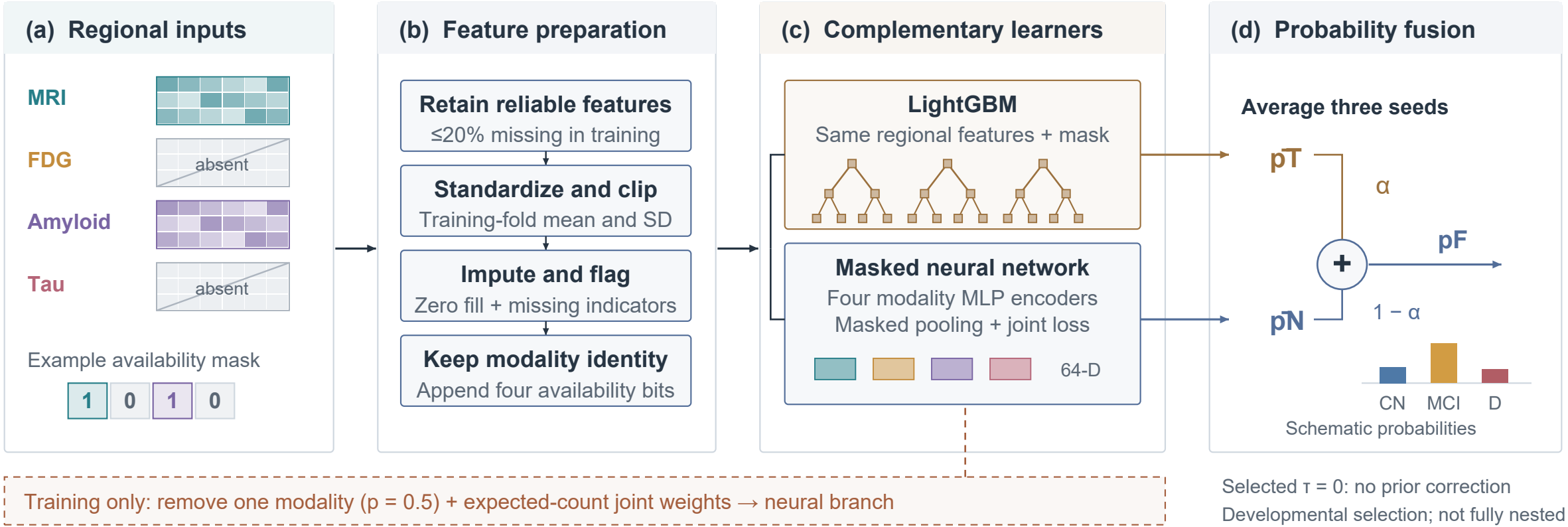

(a) Regional inputs
MRI
FDG
absent
Amyloid
Tau
absent
Example availability mask
1 0 1 0
(b) Feature preparation
Retain reliable features
≤20% missing in training
Standardize and clip
Training-fold mean and SD
Impute and flag
Zero fill + missing indicators
Keep modality identity
Append four availability bits
(c) Complementary learners
LightGBM
Same regional features + mask
Masked neural network
Four modality MLP encoders
Masked pooling + joint loss
64-D
(d) Probability fusion
Average three seeds
pT
α
pF
+
pN
1 − α
CN
MCI
D
Schematic probabilities
Training only: remove one modality (p = 0.5) + expected-count joint weights → neural branch
Selected τ = 0: no prior correction
Developmental selection; not fully nested